LeWM checkpoint reproduction

Checking the foundation before building on it

We want to establish that the released LeWorldModel (LeWM) checkpoints can complete the tasks reported in the paper using its nearby-goal evaluation. This gives our later planning and safety research a tested starting point. The test covers the complete controller: frozen model, image processing, planner and environment.

We will use the authors' available checkpoint for each environment and keep its weights fixed. The first comparison uses the original LeWM planner with the Cross-Entropy Method (CEM). Anchored Planning and our safety adaptations come after this baseline check; world-model retraining is outside this study.

Environment	Task	Paper success
TwoRoom	Navigate between rooms through a doorway	87%
Reacher	Match a target arm configuration	86%
PushT	Push a T-shaped block into position	96%
OGBench Cube	Move a single cube to its target	74%

For each case, the start and goal come from the same recorded trajectory. The goal is **25 environment actions ahead**, and the controller has **50 actions** to reach it. CEM uses 300 candidates and 30 elites. It plans five blocks of five actions and executes the 25-action plan before replanning. These scores describe nearby-goal control, rather than arbitrary full-task completion.

We will begin with the released evaluation recipe of 50 cases per environment and seed 42, saving the exact sampled cases. Additional evaluation seeds will be chosen beforehand and reported separately. Checkpoint and dataset versions, preprocessing, action normalization and success rules will be fixed. Before running, we will resolve the paper/code differences in CEM iteration counts and TwoRoom history length and record the settings we adopt.

The outcome we want is a repeatable baseline with performance consistent with the published results. We will report successes, failures and uncertainty, using an acceptable deviation chosen before seeing the outcomes. Exact equality with a rounded paper percentage is not required. Any substantial shortfall will be investigated and recorded before we build on that environment.

The final record will contain a results table, the exact configuration and case list, episode outcomes, simulator actions, model queries and elapsed time. This establishes task competence under the tested conditions; safety and longer-horizon performance require their own evaluations.